

A Convenient Schlenk Line Synthesis of $\text{Cr}_2(\text{OAc})_4 \cdot 2\text{H}_2\text{O}$

Charles G. Young

La Trobe University, Bundoora, Vic, 3083, Australia

The synthesis and manipulation of air-sensitive materials (1, 2) is an important aspect of modern inorganic chemistry. This is reflected in the publication of several air-sensitive syntheses in recent issues of *this Journal* (3–5). However, these advanced experiments demand considerable expertise and employ starting materials that are dangerous, expensive, or themselves air-sensitive. This experiment introduces simple Schlenk line techniques through the synthesis of the moderately air-sensitive compound $\text{Cr}_2(\text{OAc})_4 \cdot 2\text{H}_2\text{O}$ (OAc = acetate). The experiment offers an interesting introductory exercise that is well within the capability of students with no experience with Schlenk line techniques. Moreover, many students are familiar with $\text{Cr}_2(\text{OAc})_4 \cdot 2\text{H}_2\text{O}$ as the bonding in this and related quadruply metal–metal bonded complexes is often discussed in lectures and texts (6–8). Indeed, the determination of the structure of $\text{Cr}_2(\text{OAc})_4 \cdot 2\text{H}_2\text{O}$ (9) (see Fig. 1) provided the first evidence of strong metal–metal bonding in first-row transition metal complexes. Also, the compound is used as a starting material for the synthesis of many low-valent chromium complexes; an advanced adjunct to this experiment is the preparation of pyrophoric $\text{Cr}(\text{acac})_2$ (acac = 2,4-pentanedionate) from $\text{Cr}_2(\text{OAc})_4 \cdot 2\text{H}_2\text{O}$ (10). In their written report we ask our students to discuss the bonding and properties of $\text{Cr}_2(\text{OAc})_4 \cdot 2\text{H}_2\text{O}$, to compare their synthetic procedure with those previously published in *Inorganic Syntheses* (10–13), and, as they do not undertake its synthesis, to devise an improved synthetic procedure for the preparation of $\text{Cr}(\text{acac})_2$.

Procedure

Potential hazards attend the use of any high-vacuum technique and **protective eyewear must be worn at all times**. The experiment is performed on a small, standard Schlenk line (1). The apparatus shown in Figure 2a is assembled and clamped to a Schlenk cross (1), flask A being charged with anhydrous sodium acetate (16 g, 0.2 mol) and a magnetic stir bar. Ground glass joints and stopcocks should be carefully greased and firmly secured. A nitrogen atmosphere is created within the apparatus by performing several evacuate–backfill cycles through stopcock A. Stopcock B, which is connected to an oil bubbler, is then opened to allow a flow of nitrogen through the vessel. The flow of nitrogen allows the safe removal of cap C and the introduction of water (50 mL) onto the coarse glass frit. After replacing cap C, the water is deoxygenated by bubbling nitrogen up through it for 20 min. The nitrogen enters the apparatus at stopcock A and exits at stopcock B, as shown in Figure 2a. During this period two reagents are prepared; (1) a solution of chromium (III) chloride hexahydrate (9 g, 34 mmol) in sulfuric acid (0.2 M, 50 mL) and (2) amalgamated zinc, prepared by stirring zinc pieces (18 g) in a 0.1 M solution of mercury(II) chloride in hydrochloric acid (1 M, 150 mL) for 10 min, then washing the amalgam with water by decantation. **Caution should be exercised when handling toxic mercury compounds.** The deoxygenated water is then drawn into flask A to form a solution of sodium acetate. This is

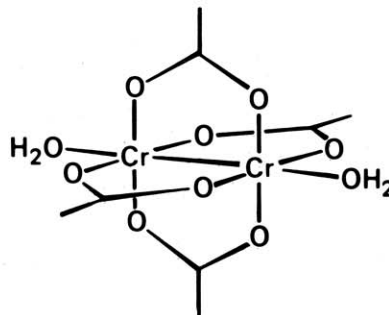


Figure 1. The structure of $\text{Cr}_2(\text{OAc})_4 \cdot 2\text{H}_2\text{O}$ (9).

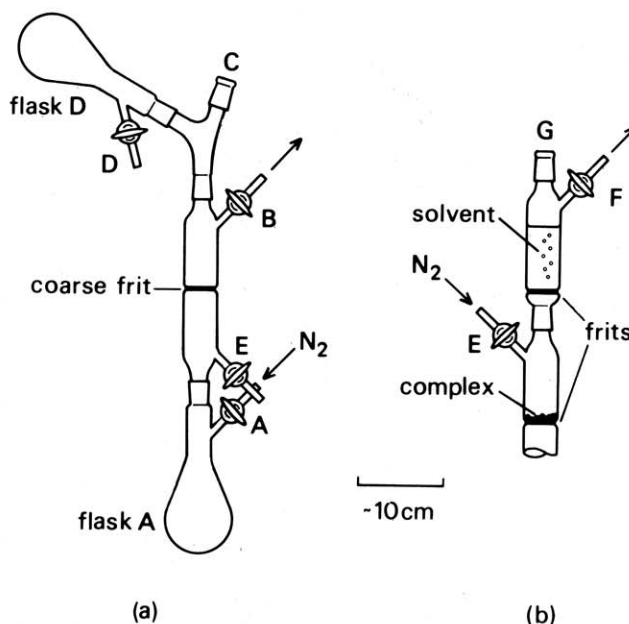


Figure 2. (a) Apparatus for the synthesis of $\text{Cr}_2(\text{OAc})_4 \cdot 2\text{H}_2\text{O}$. (b) Apparatus for the deoxygenation of washing solvents.

accomplished by closing stopcock B and applying a slight vacuum at stopcock A. After backfilling the apparatus with nitrogen and re-opening stopcock B, the amalgamated zinc and CrCl_3 solution are loaded onto the frit through port C. This mixture is deoxygenated for 30 min and the clear blue $\text{Cr}(\text{II})$ solution is drawn into flask A by the method used above for water. The precipitate of red, crystalline $\text{Cr}_2(\text{OAc})_4 \cdot 2\text{H}_2\text{O}$ is stirred for 10 min. With the aid of the rotating Schlenk cross the product is filtered by inverting the apparatus and applying a vacuum at stopcock D. The product is then washed with water (80 mL) using the following procedure. Flask A is replaced by

a deoxygenating column that has been previously filled with a nitrogen atmosphere (Fig 2b). With a flow of nitrogen through stopcocks E and F (exit to oil bubbler), water is loaded into the deoxygenating column through port G and is deoxygenated for 15 min (see Fig. 2b). The deoxygenated water is drawn onto the product by closing stopcock F and carefully applying a vacuum at stopcock E. With nitrogen entering at stopcock D careful evacuation at stopcock E allows the product to be intimately mixed with the washing solvent. The product is then freed of solvent by closing stopcock E and evacuating through stopcock D. The same procedure is used to wash the product with ethanol (120 mL). The product is then vacuum dried following the removal of the peripheral glassware from the filtration column. A simple ampule loader is used to load the dry product into ampules, which are sealed under nitrogen with a gas/oxygen torch. A typical yield is 5.5 g, 86%.

Literature Cited

1. Herzog, S.; Hehnert, J.; Luhder, K. In *Techniques in Inorganic Chemistry*; Jonassen, H. B.; Weissberger, A., Eds.; Interscience: New York, 1968; pp 119-149.
2. Shriver, D. F. *The Manipulation of Air-Sensitive Compounds*; McGraw-Hill: New York, 1969; pp 141-154.
3. Glidewell, C.; Hyde, A. R.; McKechnie, J. S.; Pogorzelec, P. J. *J. Chem. Educ.* **1985**, *62*, 534-536.
4. Villemain, D. *J. Chem. Educ.* **1987**, *64*, 183-186.
5. Lucas, C. R.; Walsh, K. A. *J. Chem. Educ.* **1987**, *64*, 265-266.
6. Cotton, F. A.; Stanley, G. G. *Inorg. Chem.* **1977**, *16*, 2668-2671.
7. Huheey, J. E. *Inorganic Chemistry*, 3rd ed.; Harper: Cambridge, 1983; pp 740-748.
8. Greenwood, N. N.; Earnshaw, A. *Chemistry of the Elements*; Pergamon: Oxford, 1984; pp 1200-1205.
9. Cotton, F. A.; DeBoer, B. G.; LaPrade, M. D.; Pipal, J. R.; Ucko, D. A. *J. Am. Chem. Soc.* **1970**, *92*, 2926-2927.
10. Ocone, L. R.; Block, B. P. *Inorg. Synth.* **1966**, *8*, 125-132.
11. Balthis, J. H., Jr.; Bailar, J. C., Jr. *Inorg. Synth.* **1939**, *1*, 122-124.
12. Hatfield, M. R. *Inorg. Synth.* **1950**, *3*, 148-150.
13. Kranz, M.; Witkowska, A. *Inorg. Synth.* **1960**, *6*, 144-146.